Multi-label Classification of Arbuscular Mycorrhizal Fungi Species

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Abstract. This work concerns arbuscular mycorrhizal fungi (AMF) prediction based on the recently constructed global AMF dataset. The global AMF dataset was generated based on dozens of various studies with different experimental design and conditions and contains batch effects. The variation partitioning analysis shows that 34% of variation in the AMF species community distribution is explained uniquely by paper label variable which is a surrogate technical variable, compared to only 3% of variation explained uniquely by environmental variables. Batch effects are taken into account by blocking by paper label when splitting the data into train, test and validation subsets, i.e., all observations of a single study fall into only one of these subsets. We handle the prediction problem in the multi-label classification setting using species presence/absence data. Multiple evaluation measures show that if one ignores batch effects one risks to obtain overly optimistic, inflated predictive performances. After controlling for batch effects, the same predictive methods show little to no generalization of AMF species presence.

1 Introduction

Arbuscular mycorrhizal fungi (AMF) are a type of mycorrhizal fungi that form symbiotic associations with the roots of most terrestrial plants [1]. A given site, defined by its pedoclimatic conditions, can host multiple AMF species forming a community of symbiotic fungi. The question we pose here is: given the pedoclimatic characteristics of a site, can we predict which AMF species are likely to be present? This presence/absence question can be addressed in the multi-label classification setting. Multi-label classification deals with the problems where an instance can be assigned multiple labels simultaneously. This problem is simpler to handle than the question of species abundances which is of compositional nature [2]: the sum of abundances for each site adds up to unity, thus the abundancies of all species should be predicted simultaneously. Standard statistical approaches can not be used in this case. On the other hand, ad-hoc approaches are challenging because of high number of species, the majority of which are rare. Dirichlet regression [3] is suitable for compositional data. But the large number of species, and the over dispersion of their abundances render this method useless.

Joint species distribution models [4] make assumptions about species distribution and model both species community associations with ecological variables,

and the species co-occurrences in the form of covariance matrices. But statistical assumptions made about complex biological systems may lead to misleading conclusions [5–7]. The goal of machine learning methods is to learn patterns from data to predict as accurately as possible the labels of new observations. General-purpose machine learning methods do not make any assumptions about the data generating distribution. Each method learns different patterns, input-output associations which may not have ecological relevance. The latter becomes even more questionable when the data is not of high quality, for instance, when it is not generated from a single well-designed experiment. This is the case with the global AMF data considered in this paper. The team [8] has created a global, publicly available AMF database from many different studies with different experimental protocols and study purposes. Machine learning methods trained on such data have high possibility to fit patterns related to study characteristics, technical artifacts or batch effects [9]. These are the patterns in the data not related to the underlying biological reality.

This work is of comparative nature. We block by paper label which is a surrogate technical variable accounting for technical particularities of a given study and split the data into train, test and validation sets. We call this blocked approach where all the observations from a single study falls into only one of these subsets. The naive approach is the one that ignores the study source of each observation and splits the data based on the species distribution. We show that this naive approach produces overly optimistic predictive results because of strong batch effects. When we take batch effects into account, generalization performances drop radically.

2 Multi-label Classification

We formalize the context of the work as follows. Let $\mathcal{X} = \mathbb{R}^d$ be an input space. We assume that the number of species is fixed and known to us (we ignore the problem of non-identified AMF species). We represent it with a finite set of labels $\mathcal{L} = \{l_1, \ldots, l_C\}$ of cardinality $C > 2$. Then, an observation $\mathbf{x} = (x_1, \ldots, x_d) \in \mathcal{X}$ represents ecological and spatial variables associated with a site, for simplicity called ecological variables in the sequel. Each observation is associated with a subset of labels in $P(\mathcal{L}) = 2^{\mathcal{L}}$. Let $\mathcal{Y} = \{0, 1\}^C$ be the presence/absence set. We map the set of labels $L(\mathbf{x}) \in P(\mathcal{L})$ associated with $\mathbf{x}$ to the corresponding element $\mathbf{y} = (y_1, \ldots, y_C) \in \mathcal{Y}$ where $y_i = 1$ if and only if $l_i \in L(\mathbf{x})$. In other words, if the species i is present in a given site, then $y_i = 1$. We do not know anything about the form of the relationship between the ecological variables $\mathbf{x}$ and the species community composition $\mathbf{y}$. The goal is to learn patterns from the sample to predict species community of new sites. Statistical learning framework handles it as follows.

Let $\mathbf{X}$ and $\mathbf{Y} = (Y_1, \ldots, Y_C)$ be random variables taking the values in $\mathcal{X}$ and $\mathcal{Y}$, respectively. We assume that a random pair $(\mathbf{X}, \mathbf{Y})$ is distributed according to some unknown joint probability distribution P on $\mathcal{X} \times \mathcal{Y}$. We denote the marginal and conditional distributions by P_X and $P_{\mathbf{Y}|X}$. The data

$D_n = \{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_n, \mathbf{y}_n)\}$, where $\mathbf{y}_i = (y_{i1}, \ldots, y_{iC})$, is the realization of n copies of $(\mathbf{X}, \mathbf{Y})$. We will use Y when referring to the binary setting. Let $\mathcal{H}$ be a space of hypotheses $h = (h_1, \ldots, h_C) : \mathcal{X} \to \mathcal{Y}$ mapping ecological/spatial variables to a set of species. For instance, $\mathcal{H}$ can be a space of linear methods or decision trees. The goal of inductive learning is to minimize over $\mathcal{H}$ the expected loss or risk:

$$R_{L,P}(h) = \mathbb{E}_{(\mathbf{X},\mathbf{Y})\sim P}[L(\mathbf{Y}, h(\mathbf{X}))],$$

i.e., output a hypothesis $h^* \in \mathcal{H}$ with low expected loss. However, since we do not know P, but have an n sample D_n, this process is usually based on the minimization of the of the empirical risk,

$$R_{L,D_n}(h) = \frac{1}{n}\sum_{i=1}^{n} L(\mathbf{y}_i, h(\mathbf{x}_i)) \tag{1}$$

plus some other regularization terms to control overfitting.

In this work we do not make any assumption about the species distribution $P_{\mathbf{Y}|X}$. In contrast, joint species distribution models make assumptions on $P_{\mathbf{Y}|X}$. For instance, species abundances are modeled by Dirichlet regression [10], species counts by Poisson log-normal [11], species presences by Bayesian logistic regression [12]. Species abundances are of compositional nature. For each location, $\mathbf{y}$ is a point in $C-1$-dimensional simplex:

$$\left\{\mathbf{y} \in [0,1]^C : \sum_{l=1}^{C} y_i = 1, y_i \geq 0 \ \forall i\right\}.$$

and thus no species in a given community can be dealt independently from the rest. Compositional data require specific, adapted statistical approaches [2] which is challenging with large number of species. Once we convert the abundance data into the presence/absence data, we get rid of the compositional structure and make the data amenable to standard statistical approaches.

The quality of data plays crucial role in statistical analysis. Having an access to high-quality data is all the more important given the huge aleatoric uncertainty related to biological/environmental processes. This uncertainty is the source of irreducible error. Unfortunately, the AMF data contain few ecological variables for reliable prediction and are affected by significant batch effects.

3 Batch Effects

Batch effects refer to unwanted technical variations in the data that are unrelated to the biological processes of interest. They are a characteristic feature of high-throughput molecular biology and genomics. Such effects may correlate with biological or environmental variables, potentially leading to misleading conclusions.

The global AMF data we consider was compiled from dozens of independent papers, each of which generated environmental DNA data using distinct experimental protocols such as different sequencing platforms, primer sets, DNA

extraction methods and so on. The authors of the compiled dataset reprocessed all raw sequencing reads from the contributing studies from scratch using the same bioinformatic pipeline with uniform quality filtering and standardized trimming procedures. For each genetic marker, sequences were restricted to a common target subregion and taxonomic assignments were made using consistent reference databases and identity thresholds. However technical differences among the original studies may still introduce systematic variation in the composition of AMF communities that does not reflect true ecological differences among sites.

To understand how much variation the study source introduces into the AMF species composition we did variation partitioning. This technique is based on redundancy analysis [13]. We used paper identifier as a technical variable. We denote it $\mathbf{X}_1$, ecological variables and geographical coordinates $\mathbf{X}_2$, and AMF species presence/absence as a response matrix $\mathbf{Y}$. The latter was Hellinger tranformed [14]. We are interested in the variation in $\mathbf{Y}$ explained by $\mathbf{X}_1$ conditioned on $\mathbf{X}_2$. This is given by the adjusted r^2 of the computation $\hat{\mathbf{Y}} = \mathbf{X}_{1,r}[\mathbf{X}_{1,r}^T\mathbf{X}_{1,r}]^{-1}\mathbf{X}_{1,r}^T\mathbf{Y}_r$ where $\mathbf{X}_{1,r} = \mathbf{X}_1 - \mathbf{X}_2[\mathbf{X}_2^T\mathbf{X}_2]^{-1}\mathbf{X}_2^T\mathbf{X}_1$ and $\mathbf{Y}_r = \mathbf{Y} - \mathbf{X}_2[\mathbf{X}_2^T\mathbf{X}_2]^{-1}\mathbf{X}_2^T\mathbf{Y}$. Similar computation holds for the variation in $\mathbf{Y}$ explained uniquely by $\mathbf{X}_2$. These computations show that the amount of variation explained uniquely by $\mathbf{X}_1$ given $\mathbf{X}_2$ is 34%, more than 10 times higher than that explained uniquely by $\mathbf{X}_2$ given $\mathbf{X}_1$ which is around 3%. This implies that predictive models fit to this dataset will learn technical artifacts instead of biological patterns. To mitigate it, we block by study source when creating train, test and validation subsets. This means all observations coming from a single study will fall in only one of these subsets. We then compare performances of learning methods trained on such subsets to those trained on the subsets created by a resampling technique not taking into consideration study source.

The next sections are dedicated to the description of predictive models, the evaluation measures, the data used, and the comparison of performances obtained based on two different resampling methods.

4 Predictive Models

AMF species prediction can be done in the binary relevance setting where each species is learned and predicted separately, and in the multi-label setting where species are learned and predicted as a community. The data splits are the same for all species irrespective of the learning setting. In the binary relevance setting we have a large choice of methods that ranges from the classical approaches such as k-nearest neighbour [15], support-vector machines [16], random forests [17], extreme gradient boosting [18], to recently developed gradient boosting models LightGBM [19], CatBoost [20], graphical neural networks [21], and auto-ML model Autogluon [22] capable of ensembling/stacking all the methods mentioned just before.

Only several methods have been developed to learn from the label dependence. The ensembles of classifier chains [23] and label powerset [24] methods are such competitive learners.

We use multiple evaluation metrics to measure their predictive performances.

5 Evaluation Measures

The multi-dimensionality of the output gives rise to an ensemble of evaluation metrics. They evaluate different aspects of classification performance and together give a broad view of it. The "few common, many rare" species distribution raises the question of suitability of particular evaluation measure.

Hamming loss is one of the most-widely used evaluation metrics in multi-label classification. This loss penalizes the misclassification for each label independently:

$$l_{HL}(\mathbf{Y}, h(X)) = \frac{1}{C}\sum_{j=1}^{C} \mathbb{1}_{\{h_j(X)\neq Y_j\}},$$

where $\mathbb{1}_{\{A\}}$ is 1 if A is true, 0 otherwise. It is not an informative measure in the context of species community prediction, since outputting only zeroes for rare species (and only ones for the most frequent ones), produces low values when averaged over the whole dataset in (1) giving an illusion of superior classification.

Another widely-used evaluation measure is the subset accuracy. The loss function related to the subset accuracy is the generalization of the traditional indicator loss function (accuracy) to the multi-label classification setting, and as such is the strictest evaluation measure since it penalizes the output of h if it does not exactly match the true labels:

$$l_{SA}(\mathbf{Y}, h(X)) = \mathbb{1}_{\{h(X)\neq\mathbf{Y}\}}.$$

In other words, if a model predicts incorrectly one species in the species community, the subset accuracy of this prediction will be zero, ignoring all correctly predicted species.

The most relevant evaluation measures for species community prediction is the family of F_1-measures. F_1-measure is used in information retrieval problems and is based on the notions of recall and precision. In the multi-label setting, we deal with 3 versions of F_1 measure.

The instance-F_1 measure is computed for each observation separately. In this case, given a site i, the recall computes the fraction of the relevant species predicted for this site:

$$\frac{\sum_{j=1}^{C} h_j(\mathbf{x}_i) y_{ij}}{\sum_{j=1}^{C} y_{ij}},$$

and the precision takes into account both relevant and irrelevant species predicted for this site:

$$\frac{\sum_{j=1}^{C} h_j(\mathbf{x}_i) y_{ij}}{\sum_{j=1}^{C} h_j(\mathbf{x}_i)}.$$

In other words, the recall answers the question, "out of all the relevant species for the given site, how many are predicted?", and the precision, "out of all

the predicted species for the given site, how many are relevant?". If all the relevant species for a given site are predicted then recall takes its maximum value. However, if there are many irrelevant species predicted for this site, it will lower the precision. The F_1-measure represents the balance between these two quantities in the form of their harmonic mean. Averaged over all samples, it is equal to

$$F_1((\mathbf{y}_i, h(\mathbf{x}_i))_{i=1}^n) = \frac{1}{n}\sum_{i=1}^{n}\frac{2\sum_{j=1}^{C} h_j(\mathbf{x}_i)y_{ij}}{\sum_{j=1}^{C} y_{ij} + \sum_{j=1}^{C} h_j(\mathbf{x}_i)}.$$

We can also compute F_1-measure for each label separately. In this case, the recall answers the question "out of all the relevant sites for the given species, in how many the species is predicted as present?" and the precision answers the question "out of all the sites for which the species is predicted as present, how many are relevant for the given species?". For species j, the recall is

$$\frac{\sum_{i=1}^{n} h_j(\mathbf{x}_i)y_{ij}}{\sum_{i=1}^{n} y_{ij}},$$

and the precision,

$$\frac{\sum_{i=1}^{n} h_j(\mathbf{x}_i)y_{ij}}{\sum_{i=1}^{n} h_j(\mathbf{x}_i)}.$$

Their harmonic mean averaged over all labels is the macro-F_1 measure:

$$\textit{macro-}F_1((\mathbf{y}_i, h(\mathbf{x}_i))_{i=1}^n) = \frac{1}{C}\sum_{j=1}^{C}\frac{2\sum_{i=1}^{n} h_j(\mathbf{x}_i)\mathbf{y}_{ij}}{\sum_{i=1}^{n} y_{ij} + \sum_{i=1}^{n} h_j(\mathbf{x}_i)}.$$

Since it is computed for each label separately, it is sensitive to the performance on rare labels.

Micro-F_1 measure aggregates the predictions from all labels and is based on the global recall and the global precision. The global recall:

$$\frac{\sum_{i=1}^{n}\sum_{j=1}^{C} h_j(\mathbf{x}_i)y_{ij}}{\sum_{i=1}^{n}\sum_{j=1}^{C} y_{ij}},$$

and the global precision,

$$\frac{\sum_{i=1}^{n}\sum_{j=1}^{C} h_j(\mathbf{x}_i)y_{ij}}{\sum_{i=1}^{n}\sum_{j=1}^{C} h_j(\mathbf{x}_i)}.$$

The harmonic mean is the micro-F_1 measure:

$$\textit{micro-}F_1((\mathbf{y}_i, h(\mathbf{x}_i))_{i=1}^n) = \frac{2\sum_{i=1}^{n}\sum_{j=1}^{C} h_j(\mathbf{x}_i)y_{ij}}{\sum_{i=1}^{n}\sum_{j=1}^{C} y_{ij} + \sum_{i=1}^{n}\sum_{j=1}^{C} h_j(\mathbf{x}_i)}.$$

To gain insight into the classification performance on each species we use Matthews correlation coefficient (MCC). It analyzes the confusion matrix as a whole and is a suitable measure in the class-imbalance setting. For species j, it is

defined as follows:

$$\mathrm{MCC}_j = \frac{\mathrm{TP}_j \cdot \mathrm{TN}_j - \mathrm{FP}_j \cdot \mathrm{FN}_j}{\sqrt{(\mathrm{TP}_j + \mathrm{FP}_j) \cdot (\mathrm{TP}_j + \mathrm{FN}_j) \cdot (\mathrm{TN}_j + \mathrm{FP}_j) \cdot (\mathrm{TN}_j + \mathrm{FN}_j)}},$$

where $\mathrm{TP}_j = \sum_{i=1}^{n} h_j(\mathbf{x}_i) y_{ij}$ is true positive, $\mathrm{TN}_j = \sum_{i=1}^{n} (1 - h_j(\mathbf{x}_i))(1 - y_{ij})$ true negative, $\mathrm{FP}_j = \sum_{i=1}^{n} (1 - y_{ij}) h_j(\mathbf{x}_i)$ false positive, and finally, $\mathrm{FN}_j = \sum_{i=1}^{n} y_{ij}(1 - h_j(\mathbf{x}_i))$ false negative.

6 Experiments

6.1 Data

The AMF data are obtained from the GlobalAMF website[1]. These data have been gathered and processed by Vetrovsky et al. [8]. The site allows one to download a file in the csv format for each species separately. Such a file for each species contains the following variables or columns: paper id, primers, longitude, latitude, continent, sample type, marker size, biome, mean annual temperature (MAT), mean annual precipitation (MAP), soil pH, year of sampling, dominant plant, target gene, sequencing platform, and abundances. This is an observational dataset collected from various studies with different experimental design, study purpose, sequencing technology etc. We remove any observations where any of the environmental variables are missing. Our downloaded data show 71 different papers/studies, 1135 unique locations, 64 species most of which have very rare occurrences.

Figure 1 shows the relative frequency of top 20 species. The most frequent species, Rhizophagus intraradices, occurs in only 35% of locations, and the majority of species have rare occurrences. Very rare species can not be used for prediction and they should be studied separately. This high sparsity of species occurrences obliges us to select only the top m most frequent species for learning. When we block by paper label, working with a large number of species makes it much more difficult to keep the label distribution similar across the splits. Note that these splits are same for all species considered. We choose $m = 10$.

For predictive learning we keep only the ecological variables. We also drop the plant variable, and use biome as its coarse surrogate. After selecting the top $m = 10$ species, we drop all lines with no species occurrences. This slightly decreases the total number of observations which remains above 1000. For this obtained dataset, the average class imbalance, defined as

$$\mathrm{avgImb}(D) = \frac{1}{C} \sum_{j=1}^{C} \frac{\max(F(j), n - F(j))}{\min(F(j), n - F(j))}$$

where $F(j) = \sum_{i=1}^{n} y_{ij}$ [25] is 4.3 and the label density, defined as

$$\mathrm{ld} = \frac{1}{nC} \sum_{i=1}^{n} \sum_{j=1}^{C} y_{ij} \in (0, 1),$$

[1] https://globalamfungi.com

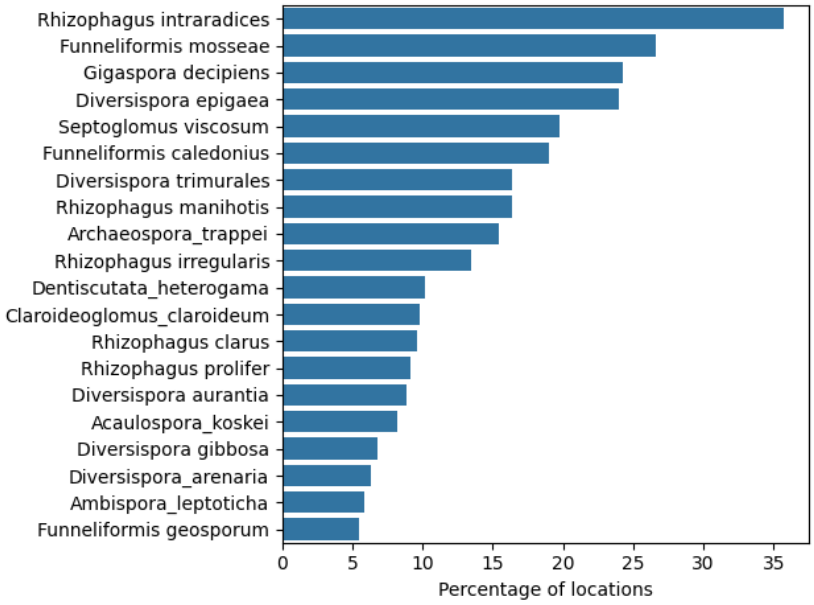


Fig. 1: Top 20 most frequent AMF species.

is quite low, 0.2.

Figure 5 shows the frequency plot of species community sizes. Around 82% of locations harbor species communities of size at most 5. Many locations are associated with one or two species, and there are single occurrences for communities of large sizes.

Table 1: Properties of the dataset: max/ls, mean/ls, and min/ls stand for the maximum, mean and minimum number of observations per label set.

n	C	avgImb	ld	max/ls	mean/ls	min/ls
1135	10	4.3	0.2	109	9	1

6.2 Model Selection

We split the data into training, validation, and test sets, comprising 60%, 20%, and 20% of the original dataset, respectively. To control for batch effects, we block by study source and randomly split the data based on the paper label to obtain approximately the mentioned percentages of observations in each of these splits. In this case, all observations of a single study falls into one of these subsets.

Keeping label distributions similar across the subsets is challenging, since all studies have different numbers of observations. In particularly, there are papers that contain a large number of only one or two species and a random split based on paper label leads inevitably to an imbalance in species frequencies across the subsets.

The naive approach ignores the study source. In this case we can use a stratified sampling approach to keep the label distributions similar across the subsets. In particular, this procedure allows the methods to learn relatively rare labels. The authors in [26] provide an iterative approach to create such subsets. The `python` implementation of this approach allows one to choose the number of labels in the label combination based on which to perform the splitting.

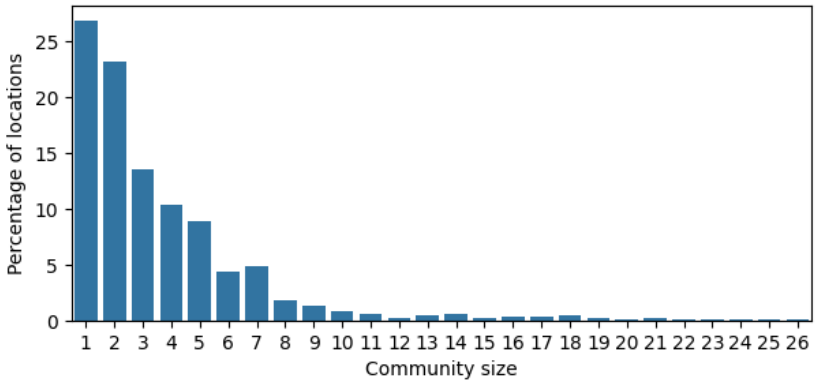


Fig. 2: Relative frequency of species community sizes over 789 unique locations.

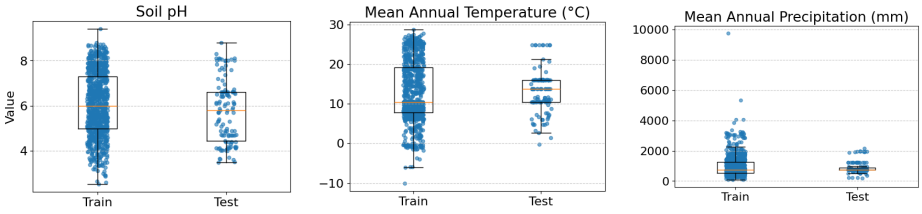


Fig. 3: Blocked splitting: distribution of environmental variables in the training and test/holdout sets.

Figure 3 shows the distribution of soil pH, MAT and MAP in the train and test sets when we block by paper label. These box plots show that the values of the environmental variables in the test set are representative of those in the train set. There is an imbalance in the land use distribution in the train and test

subsets in the blocked approach as shown in Figure 4. In particular, there are more woodland and aquatic types in the test subset. This is inevitable given the fact that not all studies consider all types of land uses.

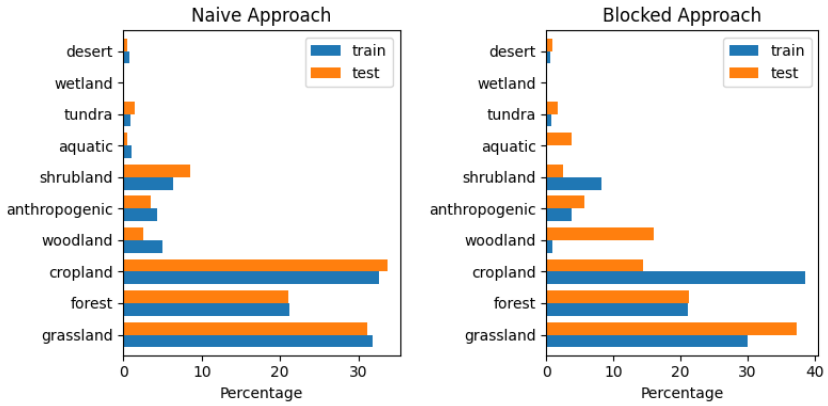


Fig. 4: Land use distribution in train and test sets obtained by naive, and blocked approaches, respectively.

6.3 Hyperparameter Tuning

The methods are trained based on the micro-F_1 measure. The hyperparameter space varies from very small (k-nearest neighbour) to highly complex (XGBoost). To this end we use `Optuna`, a hyperparameter optimization framework [27]. This framework implements grid-search, tree-structured Parzen estimator (TPE), a Bayesian optimization method for independent hyperparameter sampling, and covariance matrix adaptation evolution strategy, an evolutionary approach leveraging hyperparameter relations. We use the TPE approach useful for mixed hyperparameter setting. Grid search is used for the k-nearest neighbour.

6.4 Results

Table 2 shows predictive performances based on the naive, and the blocked approaches, respectively. Here, RF stands for random forest, GB for gradient boosting, XGB for XGBoost, ECC for ensembles of classifier chains, LP for label powerset and LGBM for LightGBM. When we block by study source, the performances drop dramatically compared to the naive approach: macro and micro-F_1 measures are below 0.5 for all methods. In the blocked approach, only autogluon which is a stacked method, demonstrates a higher predictive performance. We obtain overly optimistic predictive performances for all methods in the naive splitting setting.

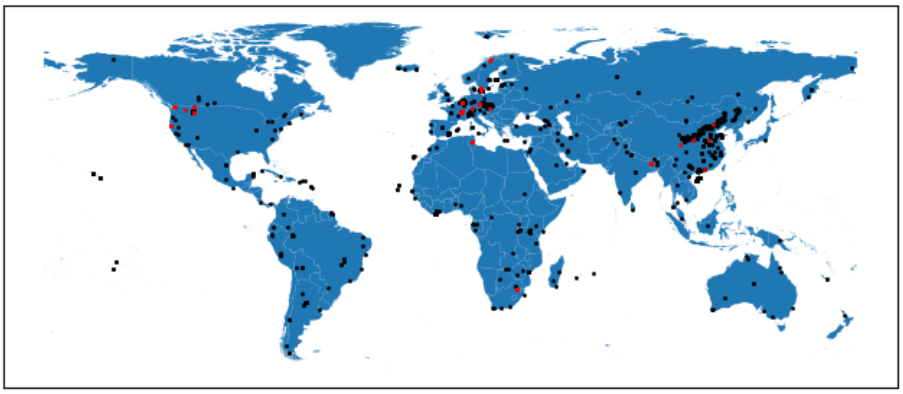


Fig. 5: Blocked splitting: locations of observations in training (black) and test/holdout (red) sets.

Table 2: Performance evaluation on the test data based on top 10 frequent species.

(a) Naive approach

Metric	knn	svc	rf	gb	xgb	ecc	lp	lgbm	autogluon
Macro-F1	0.630	0.673	0.756	0.761	0.745	0.748	0.751	0.753	0.757
Micro-F1	0.632	0.675	0.755	0.761	0.747	0.747	0.753	0.754	0.754
SA	0.276	0.342	0.442	0.447	0.367	0.427	0.457	0.432	0.417
HL	0.173	0.149	0.103	0.103	0.117	0.105	0.108	0.102	0.111

(b) Blocked approach

Metric	knn	svc	rf	gb	xgb	ecc	lp	lgbm	autogluon
Macro-F1	0.252	0.315	0.311	0.349	0.331	0.260	0.280	0.282	0.458
Micro-F1	0.299	0.371	0.415	0.422	0.383	0.348	0.324	0.366	0.477
SA	0.080	0.152	0.144	0.120	0.120	0.136	0.168	0.136	0.176
HL	0.322	0.255	0.162	0.186	0.201	0.183	0.234	0.194	0.200

Table 3 provides species-wise classification performances with respect to Matthews correlation coefficient. MCC values show that some species such as Funneliformis caledonius, Diversispora trimurales, Archaeospora trappei are not learnable in the blocked approach even though their frequencies are kept similar across the splits the values of MCC are close to zero. Negative values mean that the model prediction is the opposite of the ground truth. The blocked approach demonstrates that there is a very weak environmental signal in the data and given very limited number of pedo-climatic conditions, learnibility for the majority of species is not possible based on this dataset.

Table 3: Performance evaluation for each species independently based on the Matthews correlation coefficient.

(a) Naive approach

Species	knn	svc	rf	gb	xgb	ecc	lp	lgbm	autogluon
Rhizophagus intraradices	0.454	0.478	0.585	0.587	0.539	0.542	0.566	0.542	0.552
Funneliformis mosseae	0.401	0.529	0.607	0.609	0.631	0.658	0.623	0.671	0.662
Gigaspora decipiens	0.644	0.604	0.829	0.843	0.797	0.802	0.832	0.802	0.794
Diversispora epigaea	0.382	0.531	0.574	0.581	0.591	0.565	0.636	0.571	0.600
Septoglomus viscosum	0.525	0.548	0.578	0.571	0.590	0.598	0.548	0.673	0.592
Funneliformis caledonius	0.555	0.548	0.724	0.688	0.603	0.707	0.638	0.689	0.740
Diversispora trimurales	0.567	0.682	0.806	0.806	0.792	0.787	0.789	0.773	0.723
Rhizophagus manihotis	0.689	0.735	0.931	0.949	0.966	0.949	0.898	0.966	0.897
Archaeospora trappei	0.422	0.541	0.695	0.734	0.615	0.670	0.648	0.692	0.623
Rhizophagus irregularis	0.516	0.547	0.672	0.620	0.543	0.634	0.678	0.615	0.677

(b) Blocked approach

Species	knn	svc	rf	gb	xgb	ecc	lp	lgbm	autogluon
Rhizophagus intraradices	0.572	0.501	0.413	0.389	0.409	0.304	-0.002	0.326	0.235
Funneliformis mosseae	0.092	0.202	0.805	0.680	0.408	0.466	0.476	0.486	0.403
Gigaspora decipiens	-0.092	0.345	0.133	0.114	0.151	0.114	0.031	0.114	0.170
Diversispora epigaea	0.310	0.237	0.712	0.668	0.603	0.712	0.636	0.636	0.519
Septoglomus viscosum	0.120	0.190	0.195	0.384	0.617	0.114	0.314	0.324	1.000
Funneliformis caledonius	-0.062	0.080	0.006	0.093	0.034	-0.001	0.119	-0.008	-0.056
Diversispora trimurales	-0.161	-0.089	0.000	-0.079	0.000	-0.079	-0.079	-0.079	-0.056
Rhizophagus manihotis	-0.028	0.376	0.318	0.490	0.171	0.318	0.294	0.287	0.722
Archaeospora trappei	0.079	0.040	-0.057	-0.071	0.062	-0.073	-0.082	-0.078	0.079
Rhizophagus irregularis	0.116	0.411	-0.034	-0.034	0.166	-0.049	-0.127	-0.060	0.497

7 Conclusions

This paper considered comparative analysis of multi-label classification of the global AMF data. This is an observational dataset collected from dozens of various studies and contains batch effects. The redundancy analysis showed that a large amount of variation in the species community distribution is explained uniquely by study source variable. To control for batch effects we created splits based on the paper label, where all observations from one study fall in only one of

these splits. Predictive learning is done in the setting of multi-label classification where binary-relevance as well as the methods learning from label dependency are considered. We also considered the naive approach which ignores study source and splits the data based only on the label distribution via a stratified splitting. Only the top 10 most frequent species were considered due to a high sparsity of the majority of species. Predictive performances were evaluated based on multiple measures such as macro-, micro-F_1 subset accuracy, Hamming loss and Matthews correlation coefficient. In the naive case, these performances are overly optimistic. In particularly, macro- and micro-F_1 measures are above 0.6 for all methods. When we take batch effects into account, these F_1 measures drop below 0.5 and the methods show very weak or no generalization depending on the species. The blocked approach also showed that this dataset has very weak environmental signal for species prediction.

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